

Multimodal embedding extraction for product matching

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Product matching is a machine learning problem where we would attempt to match one product from one distribution to the same product from another distribution. The development of computer vision and natural language processing has led to the birth of pre-trained models trained on large unlabelled dataset and subsequently, fine-tuned for downstream tasks. In this paper, we propose a multimodal approach on product matching by using text and image embeddings extracted from the product information with pre-trained language and computer vision models.

CCS Concepts: • **Computing methodologies** → **Neural networks; Information extraction; Image representations.**

Additional Key Words and Phrases: product matching, multimodal embeddings, text representation model, image representation model

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1 INTRODUCTION

Product matching is a machine learning task of matching a product from one shop to another product from another shop. This task is beneficial in e-commerce where large numbers of products are presented and automatic matching algorithms are necessary instead of using human labour for matching two products.

In natural language processing, deep pre-trained representation models have been developed by first training the model on large unlabelled corpus of text using techniques such as mask language modelling and next sentence prediction [Devlin et al. 2018] and then fine tuning the model for downstream tasks. This approach is currently one of the most popular approaches in low resource training.

Similarly in computer vision, image representation models have also been trained with a large amount of images with the goal of learning features from unlabelled data. Since the breakthrough in 2012 [Hinton et al. 2012], the size of image classification models increases gradually.

In this paper, we employ a language model for extracting text embedding and a pre-trained image classification model for extracting image embedding from product offers. We subsequently concatenate both text and image embeddings to yield one single vector for one product and employ cosine similarity to compute the similarity between products' embeddings.

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2 RELATED WORK

In the early era of e-commerce, product matching was often considered as a special case of entity resolution and the task was primarily tackled using rule-based and statistical methods [Amshakala and Nedunchezian 2014; Carey et al. 2006; Cohen et al. 2003; Köpcke et al. 2012; Park et al. 2006]. Deep learning was then employed to extract better representation for the task of product matching [Barbosa 2018; Ebraheem et al. 2018; Li et al. 2020].

With the development of natural language processing (NLP), text embedding extraction is one of the most popular techniques in product matching to first extract the text embedding from product information (e.g. title, color, description) and then match these embeddings based on their similarities. [Peeters et al. 2020] proposed a fine tuning BERT approach for product matching where they used BERT to represent product text embedding information and achieved an F1 score of > 0.9 on the WDC Product Corpus for Large-Scale Product Matching. [Tracz et al. 2020] proposed an eComBERT where a BERT-based model was trained with a triplet loss objective where similar products would be grouped close to each other and different products would be pulled away from each other. [Borst et al. 2020] proposed a sentence pair classification approach also using BERT and achieved an F1 score of > 0.9 on some benchmark datasets.

Besides text embedding extraction, image embedding extraction is also used for the task of product matching. The images of products were extracted by using pre-trained computer vision models (e.g. ResNet, Inception, EfficientNet) to obtain an image embedding for each product. Similar to the text-based approach, the image-based approach also computes the similarity between products' embedding. [Gupte et al. 2021] employed ResNet-18 for their image embedding extraction whereas EfficientNet was employed in work of [Fang et al. 2021].

Multimodal embedding extraction for product matching combining both text and image extraction are also proposed recently in [Gupte et al. 2021; Ko 2021].

3 METHODOLOGY

The overall pipeline is illustrated in Figure 1. The product matching pipeline contains four main steps. First, to reduce the computational complexity, products with same brand will be grouped into brands. Second, product information will be embedded into embedding vectors from both text and image. Third, cosine similarity will be computed among all embeddings of products belong to one brand. Finally, for each product, we choose the product with maximum cosine similarity to that product to be a match. For example, to match products in brand 'quiksilver', we first group products with brand 'quiksilver' and then compute the embedding and cosine similarity for product matching. The reason behind this strategy is that both the training and test sets have a large number of samples (over 100, 000 products per set). If products are not grouped in brand,

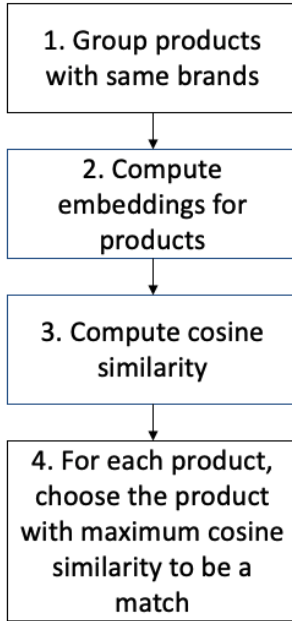


Fig. 1. The prediction pipeline for product matching

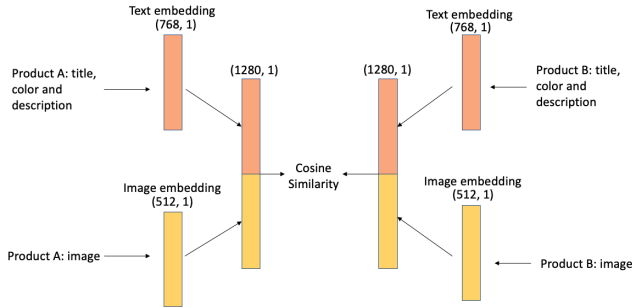


Fig. 2. Multimodal embedding extraction and cosine similarity for product matching

then 100,000x100,000 values of similarity needs to be computed (too computationally expensive).

The embedding and cosine similarity of products are shown in details in Figure 2. We first extract a text embedding from product title, color and description and an image embedding from product image. Since for each product, multiple images are presented and due to limited computational resources, we only extract embedding from the first image of each product.

3.1 Embedding extraction

For image embedding, we employ ResNet-18 proposed by [He et al. 2016] to extract features from product image URL. The output of ResNet-18 has the dimension of (512, 1). In the dataset provided by Zalando, image urls are provided and we first use HTTP request

to download the image to our memory buffer and then extract embedding using pre-trained models.

For text embedding extraction, we employ the XLM-R cross-lingual language model [Conneau et al. 2019] trained with contrastive learning via a Siamese network [Reimers and Gurevych 2019] for cross English-German languages (called Sentence BERT). Sentence BERT is based on RoBERTa [Liu et al. 2019] and is trained with a triplet loss.

3.2 Cosine similarity

To compute the similarity between two vector representations for two products, we use cosine similarity as described in a mathematical formula below:

$$\cos_sim(a, b) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (1)$$

4 EXPERIMENTS AND RESULTS

4.1 Dataset

The data we employ to test our method are from Zalando - a big e-commerce platform in Europe where training and test sets are provided. Training offer set has 102,884 product offers whereas test offer set has 106,741 product offers. For each product offer, the following information are provided: offer_id, shop, brand, color, title, description, URL to the description page, image URLs to product images, price and language. For training set, the ground truth product match is provided but for the test set, the ground truth is not provided. Therefore, all evaluation metrics in this paper is on the training set.

4.2 Experiment

For experimental design, we use built-in libraries for extracting embeddings from product information. We use the sentence_transformers library to extract text embedding and the img2vec_pytorch library to extract image embedding from product image. The dimensions of output embedding are (512, 1) and (768, 1) for image and text models, respectively.

4.3 Results

For visual inspection, we show several examples of correct matches below in Figures 3 and 4. We observe that product title, image and description of these matches are quite similar and the similarity score of these matches are high (often > 0.9).

We also an example of an incorrect match in Figure 5. Even with the same title and description the products are still different with different colors. When exploring the similarity score for this match, we see that since the titles are exactly the same, the similarity score of these products are high even when the images of products are different. This suggests that we need to further improve the factors between text and image embeddings for a product. We can also impose a threshold for similarity score so that scores larger than a threshold can be considered a match.

The evaluation result on the training set of Zalando is shown in Table 1. The F1 score we achieve on training set of Zalando challenge is 0.361.

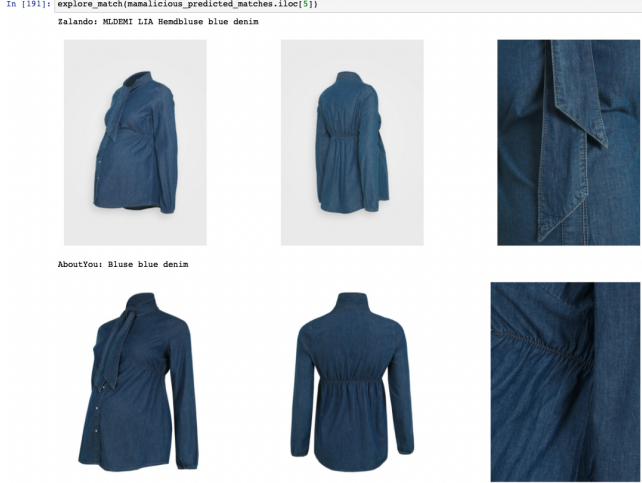


Fig. 3. An example of a correct match

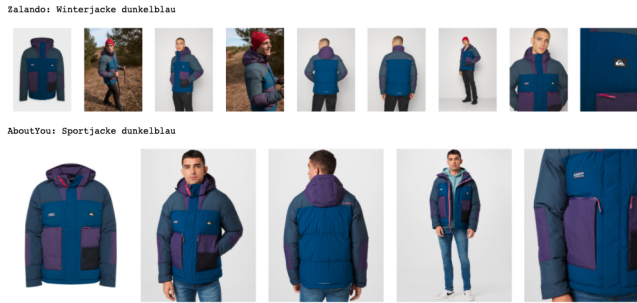


Fig. 4. An example of a correct match

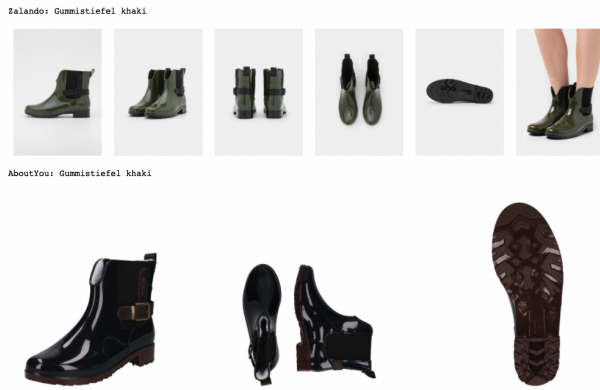


Fig. 5. An example of an incorrect match

5 CONCLUSION

In this work, we proposed a multimodal method to extract embeddings from both products' text and image information. We then compute the cosine similarity between product embeddings and for

Table 1. Product matching evaluation results

Metrics	Values
TP	4,761
TN	40,425
FP	6,385
FN	10,409
Precision	0.427
Recall	0.314
F1 score	0.361

each product, we chose another product with the highest cosine similarity to be the match. We achieved an F1 score of 0.361 on Zalando training dataset with more than 100,000 product offers. For testing, we also run our method on Zalando testing method. Since we do not have the ground truth for product matches in test dataset, we only report the F1 score on training dataset.

Future work on this project will be exploring new image representation models for the task of image embedding extraction since for the purpose of memory saving, we only use ResNet-18 for this task. We can also include different language models to experiment if other text models can work better than our current text embedding model. To further improve the matching performance, we may also apply contrastive learning to first cluster similar products and then, apply similarity matching.

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